Estimation, Likelihood, and the Bootstrap

Unit 9: Turning a Sample Into a Number You Can Defend

Stat 220 | Statistics for Data Science

Where this fits

- We know what randomness looks like and how averages behave. Now the practical question: given data, **how do you produce the number**, and how do you attach uncertainty to it?
- Sometimes the quantity is a mean and there is a formula. Usually it is a median, a ratio, a 90th percentile, or a model output, and there is not.
- This module gives you two general-purpose tools that work when formulas run out: **likelihood** and **the bootstrap**.

Principle: judge the recipe, not one result

Judge the recipe, not the number it happened to produce this time. The right question is always "if I ran this procedure on a fresh sample, how far off would it typically be?"

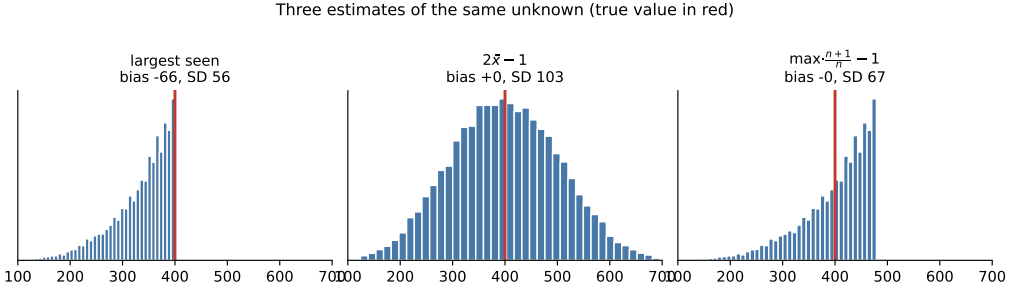
This is a real problem with a famous answer

- In World War II, Allied statisticians estimated German tank production from the serial numbers on captured tanks.
- Their estimate for 1941: about **270** tanks per month. Conventional intelligence, from spies and aerial reconnaissance, said about 1,400.
- Post-war German records: **271**.

Principle: a good estimator beats more data collection

Five serial numbers and the right formula outperformed an entire intelligence apparatus using the wrong one.

The three formulas teams usually propose



With $N = 400$ and $n = 5$: the largest ticket seen is always too low (biased by -66). Doubling the average is unbiased but wildly variable (SD 103). Correcting the maximum upward is both unbiased and tight (SD 67), and it wins the contest.

What this unit gives you

Things to know

- what an estimator is, and how to judge one by **bias**, **variance**, and MSE
- why the best estimator is often a slightly biased one
- shrinkage, and why every leaderboard needs it
- what a likelihood is, and what maximum likelihood actually maximizes
- why the curvature of the likelihood is the precision of the estimate
- how the bootstrap manufactures a sampling distribution from one sample

Mistakes to stop making

- chasing unbiasedness at the cost of enormous variance
- reporting an estimate with no standard error
- confusing the SD of the data with the SE of the estimate
- bootstrapping rows when the independent unit is a user, store, or week
- bootstrapping maxima or extreme quantiles
- ranking groups of very different sizes without shrinkage

Three ways an estimate can be wrong

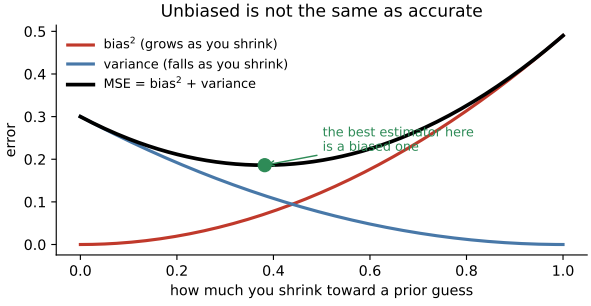
$$\underbrace{E[(\hat{\theta} - \theta)^2]}_{\text{MSE}} = \underbrace{(E[\hat{\theta}] - \theta)^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{\theta})}_{\text{variance}}$$

- **Bias:** on average, does it land on the truth?
- **Variance:** how much does it bounce from sample to sample?
- **MSE** combines them, and it is the number you should actually care about.

Principle: aim for accurate, not merely honest

An estimator that is right on average but wildly variable is useless in the single sample you actually have. You get one draw.

Why a biased estimator can win



Pulling an estimate partway toward a sensible prior guess (**shrinkage**) buys a large drop in variance for a small amount of bias.

- Total error falls.
- This is the same tradeoff behind regularization, hierarchical models, and every "smoothed" metric you have seen.

The 3-for-3 salesperson

Setup. A new rep closed all three of her first three deals. The unbiased estimate of her close rate is 100%.

- Nobody believes 100%. With three attempts, the estimate has enormous variance.
- A better estimate shrinks toward the company average: for instance, add the equivalent of a few "average" attempts before computing the rate.
- That is exactly how batting averages, restaurant ratings, and seller scores are computed in practice.

In practice: a favorite question

"Which product has the best rating: 5.0 from 2 reviews, or 4.6 from 800?" The answer requires shrinkage, and saying so out loud is the point.

Your turn #1

Your turn: design the leaderboard

You run a marketplace. You must rank 10,000 sellers by customer rating. Some have 3 reviews, some have 3,000.

With a neighbor: what goes wrong if you rank by raw average? Propose a concrete fix, and say what the fix costs you.

Your turn #1: answer

- Raw averages put brand-new sellers with two five-star reviews at the top of every page. The leaderboard becomes a list of small samples, not good sellers.
- Fix: shrink each seller's average toward the site-wide average, with the weight depending on the review count. A seller with 3 reviews barely moves from the site average. A seller with 3,000 keeps their own.
- The cost: genuinely excellent new sellers are understated for a while. That is the bias you are accepting in exchange for a stable, non-gameable ranking.

Principle: small samples must be handled, not trusted

Any ranking built from unequal sample sizes needs shrinkage. Without it, you are ranking noise.

Where do estimators come from?

Two general recipes:

- **Method of moments:** choose the parameter that makes the model's mean match the sample mean. Simple, intuitive, occasionally poor.
- **Maximum likelihood:** choose the parameter under which the data you actually observed is most probable.

Principle: maximum likelihood in one sentence

"Given this model, which parameter value would have made what I saw the least surprising?"

Method of moments

The simplest recipe there is: *make the model's average match the data's average*, then solve.

- Data: 40 support calls per day on average. Model: Poisson(λ). Poisson's mean is λ, so set $\hat{\lambda} = 40$. Done.
- Data: mean 0.30 conversion. Model: Bernoulli(p), whose mean is p, so $\hat{p} = 0.30$.
- It is quick, it is intuitive, and it often lands on the same answer as maximum likelihood.

Trap: where it falls apart

Method of moments uses only the mean (and maybe the variance). It throws away the rest of the data's shape, which is exactly the information you need when the model has several parameters or the data is censored.

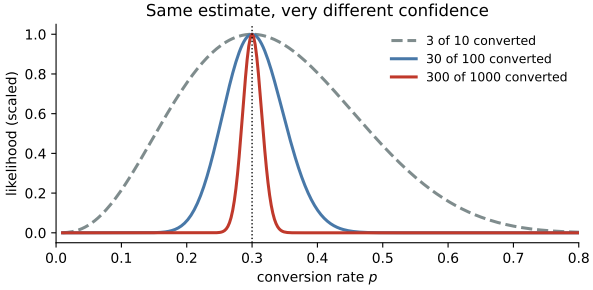
Likelihood, concretely

You email 100 customers and 30 convert. If the true rate is p, the probability of seeing exactly this data is

$$L(p) = \binom{100}{30} p^{30}(1-p)^{70}.$$

- As a function of p, this is the **likelihood**. It is not a probability distribution over p. It ranks parameter values by how well they explain the data.
- It is maximized at $\hat{p} = 0.30$, the sample proportion. Maximum likelihood usually recovers the obvious estimator, and then keeps working when nothing is obvious.

The curvature is the confidence



All three datasets estimate $\hat{p} = 0.30$. They are not equally convincing.

- 3 of 10: a broad, flat curve. Many values explain the data nearly as well.
- 300 of 1000: a sharp spike. Only a narrow range is compatible.

Sharp curve, small standard error. That is where SEs come from in general.

Why this matters beyond the formula

- Every model you will fit later (logistic regression, Poisson regression, mixed models) is fit by maximizing a likelihood.
- Every standard error those models print comes from the curvature of that likelihood at its peak.
- So "the model is very uncertain about this coefficient" means "the likelihood is flat in that direction," which usually means the data contains little information about it.

Trap: a flat likelihood still prints a number

Software always prints a number. If two predictors are nearly identical, the likelihood is flat along a ridge, and the individual coefficients are nearly arbitrary even though the fit is fine.

The problem the bootstrap solves

- We know the standard error of a *mean*: $\sigma/\sqrt{n}$.
- What is the standard error of a **median**? A 90th percentile? A ratio of two medians? The difference in churn between two segments after adjusting for tenure?
- Formulas exist for a few of these and nobody remembers them. The bootstrap needs none.

Principle: the bootstrap idea

The sample is your best available picture of the population. So resample *from the sample*, with replacement, and watch how much your estimate moves.

The bootstrap recipe

1. Draw a new sample of size n from your data, **with replacement**.
2. Compute your statistic on it. Anything at all: median, ratio, model coefficient.
3. Repeat a few thousand times.
4. The spread of those values is the standard error. The middle 95% of them is a confidence interval.

Four lines of code, and it works for statistics that have no formula.

The bootstrap, as code

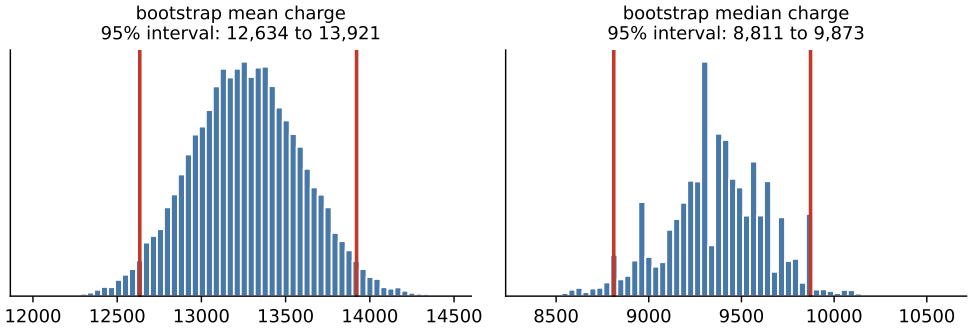
```
boot = []
for i in range(5000):
resample = sample_with_replacement(data, size=len(data))
boot.append(your_statistic(resample))

se = std(boot)
ci = percentile(boot, [2.5, 97.5])
```

- `your_statistic` can be anything: a median, a 90th percentile, a ratio of two medians, an R^2, a model coefficient.
- Nothing in there assumes a distribution, requires a formula, or needs a derivation.

Bootstrapping real data

Resampling the data gives an interval with no formula at all



Medical charges for 1,338 people. The mean and the median have very different intervals, and the median's is far tighter, because it ignores the tail. Which one you report depends on the question, not on which looks better.

Your turn #2

Your turn: which one do you report?

Your company's charges data gives a bootstrap 95% interval of roughly \$12,600 to \$13,900 for the **mean**, and \$8,800 to \$9,900 for the **median**.

With a neighbor: give one business question where the mean is the right answer, and one where the median is, and say why reporting only one could mislead.

Your turn #2: answer

- **Mean:** anything about totals. Budgeting the year's payouts requires the mean, because the total is the mean times the number of people.
- **Median:** anything about a typical person. "What does a customer like me pay?" is a median question.
- Reporting only the mean makes the typical experience sound worse than it is. Reporting only the median makes the budget look far too small.

Principle: the statistic follows the decision

Choose the summary from what someone will *do* with it, then bootstrap whatever you chose.

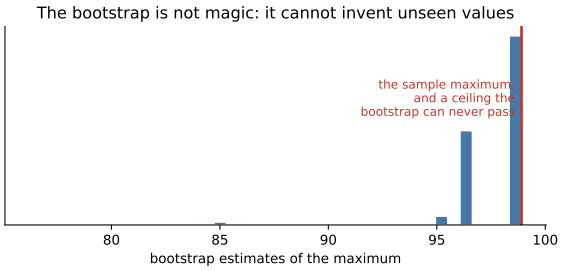
Three ways to get a standard error

Method	How	Use it when
Formula	$s/\sqrt{n}$ and its relatives	the statistic is a mean or a slope and the assumptions hold
Likelihood curvature	how sharply the log-likelihood peaks	you fit a model by maximum likelihood, which is what software reports
Bootstrap	resample and look at the spread	anything else, or when you do not trust the formula's assumptions

Principle: they should agree, and it is worth checking

On a well-behaved mean, all three land in the same place. When the bootstrap disagrees with the formula, believe the bootstrap and go find out which assumption broke.

Where the bootstrap fails



Resampling can never produce a value you did not observe, so it cannot see past the largest point in your data.

- Fails for maxima, minima, and extreme quantiles.
- Struggles with very small n, since it can only reshuffle what little you have.
- Breaks entirely on dependent data unless you resample *clusters*.

Your turn #3

Your turn: what is one observation?

For each dataset, say what you would resample, and what your real n is:

1. Mean session length, from 100,000 page views by 6,000 visitors.
2. Mean daily sales, from 365 consecutive days at one store.
3. The district average, from 900 students in 30 classrooms.

Your turn #3: answer

- **Visitors**, not page views. Real n is about 6,000, and resampling page views would understate the SE by roughly a factor of four.
- **Blocks of consecutive days**, not individual days. Sales are autocorrelated, so a day is not an independent draw, so resample contiguous stretches to keep the dependence.
- **Classrooms**, not students. Real n is closer to 30 than 900, because students in a room share a teacher, a schedule, and a neighborhood.

Principle: the question that comes before the code

"What is one independent observation?" If you cannot answer it, no interval you compute means anything, however sophisticated the method.

Bootstrapping dependent data, correctly

- If you have 100,000 sessions from 4,000 users, resampling *sessions* pretends you have 100,000 independent pieces of information. Your interval will be far too narrow.
- Resample **users** instead, keeping all of a user's sessions together. Same idea for stores, classrooms, or hospitals.
- For time series, resample contiguous *blocks* rather than individual days, to preserve the dependence.

Trap: the most common bootstrap error in industry

Bootstrapping rows when the independent unit is something coarser. The code runs, the interval is tight, and it is wrong.

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM for an interval

We told it: "I have 500 session times and want a 95% confidence interval for the *median* session time."

It answered: "Compute the median, then use $\bar{x} \pm 1.96\, s/\sqrt{n}$ with the sample standard deviation. That gives a 95% confidence interval for the median."

Mistake, or no mistake?

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Mistake, or no mistake?

Trap: verdict: mistake

That formula is the interval for the *mean*, and on skewed session times the mean and median are far apart. The clean answer is to bootstrap the median directly, which needs no formula at all.

LLM check: mistake or not? (2 of 2)

LLM check: same territory, a different answer

We told it: "We have 80,000 support tickets from 5,000 customers. Bootstrap the average resolution time."

It answered: "I would resample *customers*, not tickets, keeping each customer's tickets together. Tickets from the same customer are correlated, so resampling tickets would treat 80,000 rows as independent and produce an interval that is much too narrow."

Mistake, or no mistake?

LLM check: mistake or not? (2 of 2)

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Mistake, or no mistake?

Principle: verdict: no mistake

It identified the independent unit before writing any code. That single habit separates a defensible interval from a confident-looking one.

How to estimate anything

1. **Write the target in words.** "Median revenue per active user in Q3," not "revenue."
2. **Identify the independent unit.** Users? Stores? Weeks? This decides your real n.
3. **Choose the statistic from the decision**, not from convenience.
4. **Compute the estimate**, then immediately ask how much it would move on a fresh sample.
5. **Get the uncertainty**: a formula if one applies, otherwise a bootstrap over the independent unit.
6. **Consider shrinkage** if any group has a small sample.
7. **Report estimate, interval, and the assumption you are least sure about.**

Scenario: the regional conversion leaderboard

Setup. Marketing ranks 40 regions by conversion rate. The top region converts at 22% against a company average of 9%. They want to copy that region's playbook everywhere.

- First question: how many visitors did that region have? If the answer is 45, its 22% is one lucky week.
- Second: with 40 regions, the maximum of 40 noisy estimates is high *by construction*, even if all regions are identical.
- What to do: shrink each region's rate toward the company average, bootstrap the intervals, and check whether next quarter's data agrees.

You may be asked: how to answer

"The winner of a leaderboard is a biased estimate of the best performer. I would shrink, then validate on new data before rolling out a playbook."

Estimating anything, with an interval

1. **Write the target in words and units**, for example "median revenue per active user."
2. **Identify the independent unit**: users, stores, or weeks. This sets your real n.
3. **Choose the statistic from the decision**, not from convenience.
4. **Compute the estimate.**
5. **Resample with replacement** a few thousand times, recomputing the statistic each time.
6. **Take the SD of those values** for the standard error and the 2.5 and 97.5 percentiles for the interval.
7. **Shrink toward a sensible prior** if any group has a small sample, then report estimate, interval, and the shakiest assumption.

The traps

Trap: the ones that bite most often

- Treating unbiasedness as the only virtue and ignoring variance.
- Reporting a point estimate with no standard error or interval.
- Confusing the SD of the data with the SE of the estimate.
- Bootstrapping rows when the independent unit is a user, store, or week.
- Bootstrapping maxima or extreme quantiles.
- Ranking groups of very different sizes without shrinkage.

Ten questions to be able to answer

1. What makes one estimator better than another? Define bias, variance, and MSE.
2. Give an example where a biased estimator is the better choice.
3. Explain maximum likelihood in plain English.
4. Why does the shape of the likelihood tell you the standard error?
5. How would you get a confidence interval for a 90th percentile?
6. Explain the bootstrap to a product manager in thirty seconds.
7. When does the bootstrap fail, and what would you do instead?
8. We have 80,000 rows from 5,000 customers. What do you resample?
9. A new rep is 3-for-3. What is your estimate of her close rate?
10. How would you build a leaderboard that new sellers cannot game with two reviews?

Mastery checklist

You have this unit when you can, from memory:

- decompose an estimator's error into bias and variance and argue about the tradeoff
- explain why shrinkage improves small-sample estimates
- define the likelihood and say what maximum likelihood chooses
- connect the curvature of the likelihood to the standard error
- run and explain a bootstrap, including the choice of resampling unit
- name three situations where the bootstrap should not be trusted

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Resample rows with replacement	`rng.integers(0, n, size=(B, n))`
Bootstrap SE and percentile interval	`boot.std()`, `np.percentile(boot, [2.5, 97.5])`
Bootstrap a median, a percentile, or a ratio	the same loop, a different statistic
Resample clusters instead of rows	sample the group labels, then pool their rows
Maximum likelihood for a simple model	`scipy.optimize.minimize` on the negative log-likelihood
Bias, variance, and MSE of an estimator	simulate from a known truth and compare
Shrunken group estimates	`(sum + k*overall) / (n + k)`

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where we go next

- You can now produce an estimate and an honest interval for almost any quantity, with or without a formula. That closes out the foundations.
- **Unit 2** puts the machinery to work on a relationship rather than a single number: regression, where the estimate is a slope and every idea from Units 1 through 4 comes back wearing new clothes.

Principle: the through-line

Estimation is not about getting a number. It is about knowing how much that number would have changed if the world had rolled the dice again.